DO NOW — RECALL

# Recursion: write and debug algorithms

Name: _____ Class: _____ Date: _____

**1. SumTo(n) returns 1 when n=1; otherwise it returns n+SumTo(n-1). For SumTo(4), identify the call that returns first and the call that returns last.**

**2. Why must a recursive function contain a stopping condition?**

**3. A loop repeatedly halves n starting at 40 while n > 1 using integer division. List the next three n values.**

**4. A function PowerTwo(n) returns 1 when n=0, otherwise 2*PowerTwo(n-1). What does PowerTwo(3) return?**

**DO NOW — CHECK / OBJECTIVES**

# Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

### 1. Model answer

First: SumTo(1). Last: SumTo(4). The nested calls finish before their callers return.

### 2. Model answer

Without a reachable base case, calls continue and the call stack grows until failure.

### 3. Model answer

20, 10, 5.

### 4. Model answer

PowerTwo(0)=1, PowerTwo(1)=2, PowerTwo(2)=4 and PowerTwo(3)=8.

### Vocabulary — know these terms

termination — Reaching a condition where the algorithm stops.

recursive step — Processing part of a problem before calling the function on a simpler input.

precondition — An assumption or input rule required for an algorithm.

stack overflow — A failure that can occur when too many active calls consume stack space.

integer division — Division discarding any fractional part of the result.

NEW LEARNING

# Read, annotate and explain

### Find the smallest problem

A positive single-digit number needs one digit. So n < 10 is our base case.

### Make progress

For n ≥ 10, integer division by 10 makes the number shorter: 4827 → 482 → 48 → 4.

### Count on the way back

4 returns 1. Then 48 returns 2; 482 returns 3; 4827 returns 4.

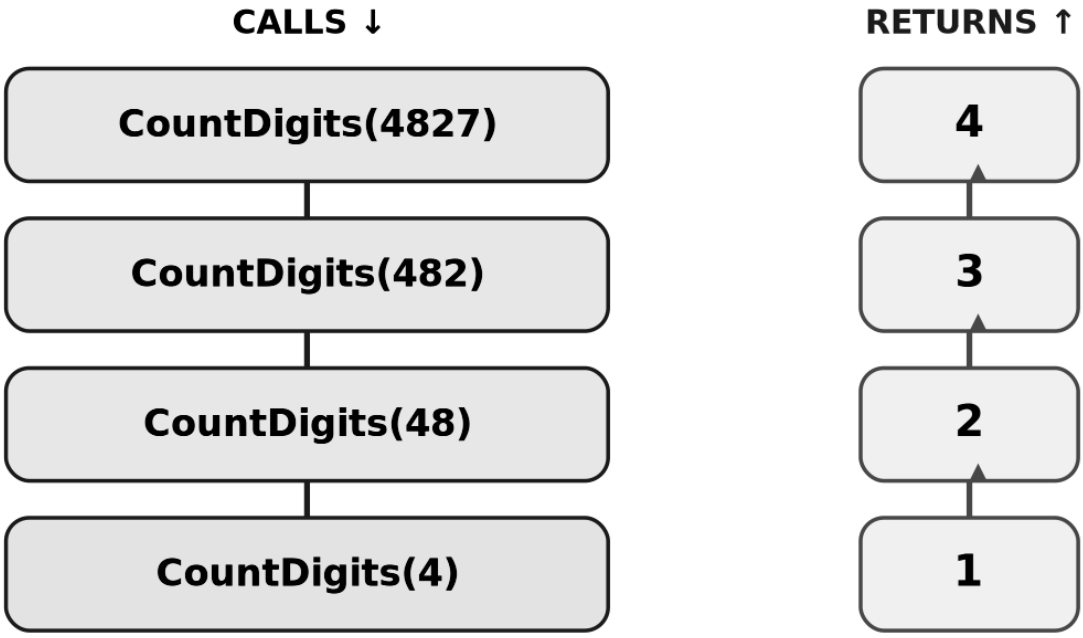
### Explain one step in your own words, using the correct technical terms.

# New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

### What is the program actually doing?

Assume n is positive . Integer division by 10 removes the final digit at each call.




Recursive calls moving downward and return values moving upwards. CountDigits(4827), CountDigits(482), CountDigits(48), CountDigits(4).

I DO — WORKED EXAMPLE

# Follow the worked method

### Model CountDigits(582). Why is integer division essential?

### Calls

582 → 58 → 5. Each integer division by 10 discards the last digit.

### Base case

5 is a single digit, so CountDigits(5) returns 1.

### Returns

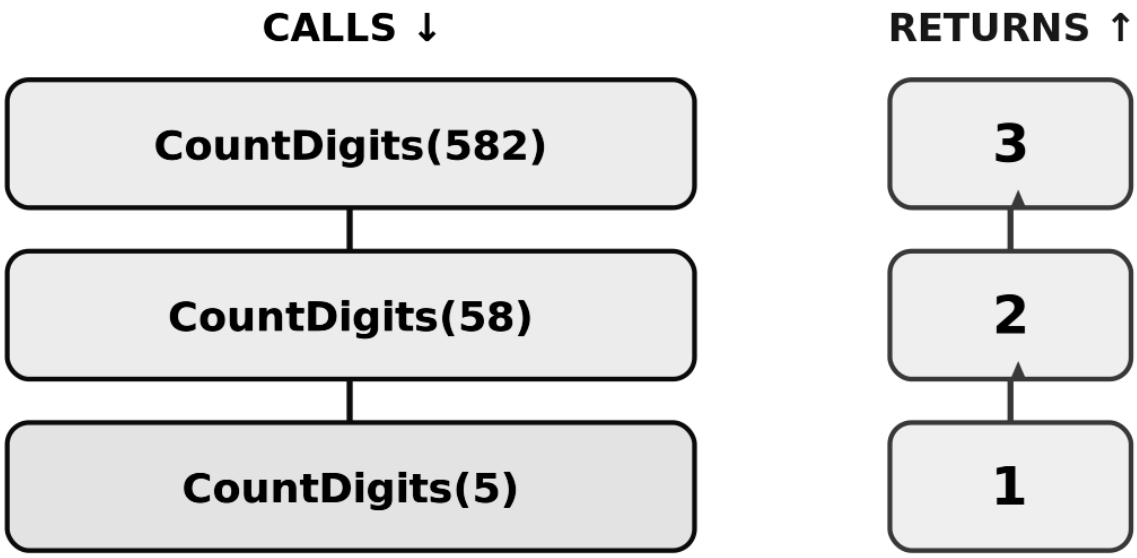
CountDigits(58)=2; CountDigits(582)=3. The original number has 3 digits.

# I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

### Model CountDigits(582). Why is integer division essential?

CountDigits(582) → CountDigits(58) → CountDigits(5)




Call trace and return trace: CountDigits(582); CountDigits(58); CountDigits(5). Return values: 3, 2, 1.

WE DO — GUIDED PRACTICE

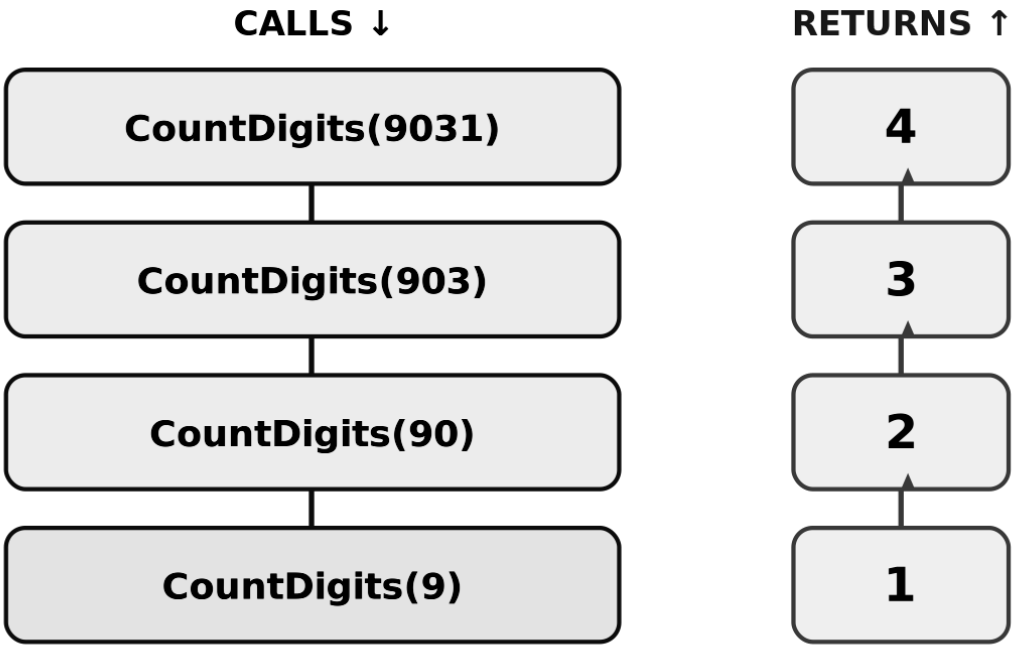
# Try → Discuss → Improve

# We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

**Trace CountDigits(9031). Predict each call and returned value.**

9031 → 903 → 90 → 9




Call trace and return trace: CountDigits(9031); CountDigits(903); CountDigits(90); CountDigits(9). Return values: 4, 3, 2, 1.

# WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

### Guided 1: Calls 9031, 903, 90, 9.

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

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YOU DO — INDEPENDENT PRACTICE

# Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

### Question 1

Trace CountDigits(67): write the call sequence and final value.

### Question 2

A pupil writes return CountDigits(n) + 1; for the recursive step. Correct the line and explain the error.

YOU DO — INDEPENDENT PRACTICE

# Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

### Question 3

Extend CountDigits for n = 0 and negative integers without overflowing on the minimum C# int value.

EXIT TICKET / REFLECTION

# Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

**An algorithm is recursive whenever it uses a loop.**

**How confident are you? Circle one: Confident / Getting there / Need help**

Explain your choice, identifying one concept you can now use independently.

**GREEN PEN — Correct and explain**

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.

# WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

### Guided 1 — original CHECK

Base case

9 is a single digit so returns 1.

Returns

9 returns 1; 90 returns 2; 903 returns 3; 9031 returns 4.

IMPROVE: Explain one correction to your own reasoning.